

1 Solution — GIAM 4.1

Problem 9

1.1 Problem

How many 8-element subsets are there in $\mathcal{P}(\{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p\})$?

1.2 Answer

$$\binom{16}{8} = 12\,870$$

1.3 Why

The set has 16 elements, and we are counting subsets of size 8.

Building an 8-element subset one letter at a time, there are 16 choices for the first, 15 for the second, and so on down to 9 for the eighth:

$$16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 = 518\,918\,400.$$

*But this counts **ordered** selections, and a set does not record order. Each 8-element subset arises from $8! = 40\,320$ different orderings of the same eight letters, so every subset has been counted $8!$ times. Dividing,*

$$\binom{16}{8} = \frac{518\,918\,400}{40\,320} = 12\,870.$$

This is exactly the correction made in the previous problem, where 5×4 ordered pairs became 10 doubletons after dividing by $2!$. Here the divisor is $8!$ rather than $2!$.

1.4 The Picture

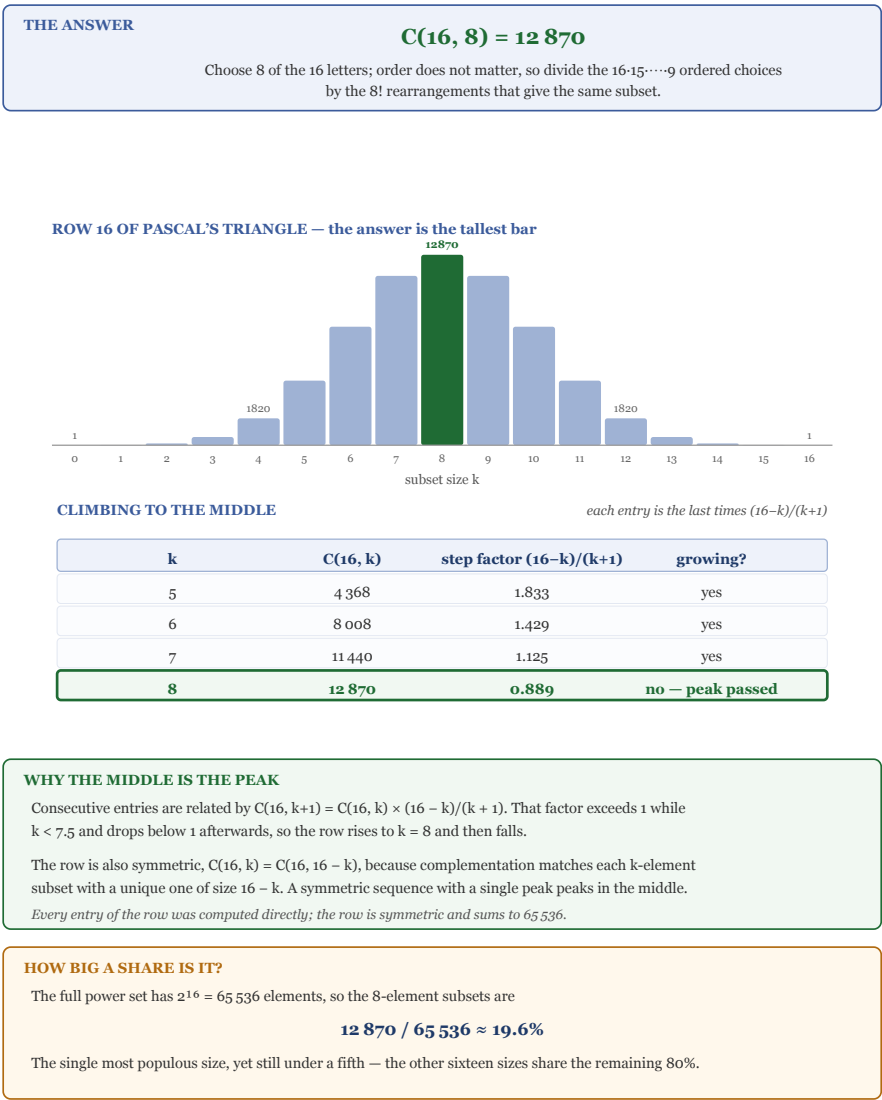


Figure 1.1: A diagram in four parts. The first gives the answer: C of 16 and 8 equals 12870. Choose 8 of the 16 letters; order does not matter, so divide the 16 times 15 and so on down to 9 ordered choices by the 8 factorial rearrangements that give the same subset. The second part draws row 16 of Pascal’s triangle as a bar chart, with one bar per subset size from 0 to 16. The bars rise from height 1 at size 0 through 16, 120, 560, 1820, 4368, 8008 and 11440, peak at 12870 for size 8 where the bar is coloured green, then fall symmetrically back to 1 at size 16. Selected values are labelled above their bars. A table then shows the climb to the middle, listing k equal to 5, 6, 7 and 8 with the values 4368, 8008, 11440 and 12870, alongside the step factor 16 minus k over k plus 1, which is 1.833, 1.429, 1.125 and 0.889 respectively, and whether the sequence is still growing — yes for the first three, and no, peak passed, at k equals 8. A band below explains why the middle is the

peak: consecutive entries satisfy C of 16 and k plus 1 equals C of 16 and k times the quantity 16 minus k over k plus 1, and that factor exceeds 1 while k is below 7.5 and drops below 1 afterwards, so the row rises to k equals 8 and then falls. The row is also symmetric because complementation matches each k -element subset with a unique one of size 16 minus k , and a symmetric sequence with a single peak peaks in the middle. Every entry was computed directly; the row is symmetric and sums to 65536. A final band asks how big a share this is: the full power set has 2 to the sixteenth, or 65536, elements, so the 8-element subsets are 12870 out of 65536, about 19.6 percent — the single most populous size, yet still under a fifth, with the other sixteen sizes sharing the remaining 80 percent.

1.5 Remark — This Is the Largest Entry in Its Row

Row 16 of Pascal’s triangle, computed in full:

1, 16, 120, 560, 1820, 4368, 8008, 11440, **12870**, 11440, 8008, 4368, 1820, 560, 120, 16, 1

The answer sits at the centre, and it is the biggest number in the row. Two facts explain that.

The row is symmetric. $\binom{16}{k} = \binom{16}{16-k}$, because complementation $T \mapsto S \setminus T$ pairs each k -element subset with exactly one of size $16 - k$. Verified: the list reads the same backwards.

The row has a single peak. Consecutive entries satisfy

$$\binom{16}{k+1} = \binom{16}{k} \cdot \frac{16-k}{k+1},$$

and the factor $\frac{16-k}{k+1}$ exceeds 1 precisely when $k < 7.5$. So the entries strictly increase up to $k = 8$ and strictly decrease after:

k	$\binom{16}{k}$	step factor
6	8008	1.429
7	11440	1.125
8	12870	0.889

A symmetric sequence with one peak must peak at the middle, which is why the answer to “how many subsets of size 8” is the largest such count available.

1.6 Remark — How Much of the Power Set Is This?

The whole power set has $2^{16} = 65\,536$ elements, so the 8-element subsets make up

$$\frac{12\,870}{65\,536} \approx 19.64\%.$$

Under a fifth — which is worth noticing, because “most common size” is easily misread as “most of them”. The row sums to **65 536** exactly, confirming that the seventeen sizes account for everything, with the remaining sixteen sizes sharing about 80%.

The share also shrinks as the set grows. The central binomial coefficient satisfies

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}},$$

so its share of $2^{2n} = 4^n$ is roughly $1/\sqrt{\pi n}$, which tends to 0. At $n = 8$ the approximation gives **13 072.5** against the true **12 870** — within 1.6% — and predicts a share of $1/\sqrt{8\pi} \approx 20.0\%$, close to the observed 19.64%.

1.7 Remark — Why Not Just List Them

The previous problem listed all 5 singletons and all 10 doubletons. That is not an option here, and the jump is instructive.

singletons of a 5-set	5
doubletons of a 5-set	10
8-element subsets of a 16-set	12 870

Listing 12 870 subsets would fill many pages and demonstrate nothing. This is the point at which a counting *formula* stops being a convenience and becomes the only practical method — and it is why the binomial coefficient is worth having as an object in its own right rather than as shorthand for a tally.

Note also that the formula answers the question without ever constructing a single subset. That distinguishes it from the exhaustive arguments of Chapter 3, where the objects had to be produced and inspected one by one.

1.8 Remark — Three Ways to Compute It

The same number can be reached by three routes, which is a useful check.

By the formula. $\binom{16}{8} = \frac{16!}{8!8!} = \frac{518\,918\,400}{40\,320} = 12\,870.$

By cancelling first. Writing the quotient before multiplying keeps the numbers small:

$$\frac{16}{8} \cdot \frac{15}{7} \cdot \frac{14}{6} \cdot \frac{13}{5} \cdot \frac{12}{4} \cdot \frac{11}{3} \cdot \frac{10}{2} \cdot \frac{9}{1},$$

which is how one would do it by hand — $16!$ itself is over 2×10^{13} and needlessly large.

By Pascal's rule. $\binom{16}{8} = \binom{15}{7} + \binom{15}{8} = 6435 + 6435 = 12\,870$, splitting on whether a fixed letter is in the subset. The two halves are equal here, which is another way of seeing the central symmetry.

All three agree, and the row was generated independently to confirm the value.